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SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

Explore the artistic skills of animals, the outrageous plans to halt climate change, the out of this world experiments on board the International Space Station and the various attempts to bend light itself.

Spooky action at a distance confirmed by quasars

Quantum entanglement is a phenomenon where the properties of two particles are linked to each other, no matter what the distance between them. Einstein called the phenomenon "spooky action at a distance". A team of researchers at MIT found correlations between 30,000 photons released by quasars 7.8 and 12.2 billion years ago, long before the experiments were concieved.
https://dgit.in/qsaad

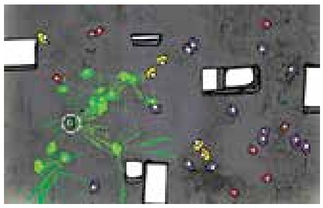
WHAT'S NEW

Laughing gas may have kickstarted life on Earth

In the very early days of Earth's history, the Sun shone much more dimly than it does today. Despite that, a greenhouse gas effect kept the Earth warm and cozy. Carl Sagan referred to this condition as the "the Faint Young Sun Paradox". While models have shown that there were high amounts of carbon dioxide and methane in the atmosphere during the Proterozoic Eon, they together do not account for how warm the planet actually was. According to new research this gas could be nitrous oxide, better known as laughing gas. The study suggests that the quantity of nitrous oxide was sufficient to keep the Earth warm. It is also very possible that the most ancient life forms were actually breathing nitrous oxide rather than oxygen. There are similarities between the enzymes that organisms use to breathe nitrous oxide and oxygen.
https://dgit.in/n2Olife

Using mathematics to understand disease spread

Current scientific models of the behavior of pathogens within the host communities do not take one important factor into account - social behavior. The relationship between the spread of diseases and social factors is not understood very well. Human behavior may change in response to the outbreak of a disease, and this is not factored into the understanding of how the disease emerges and spreads. For example, humans may start wearing facemasks, and subsequently stop wearing them too soon. The disease reacts to the human responses, and the human attempts at controlling the spread of the disease. The response by humans is continuous through the duration of the outbreak. New models help healthcare workers come up with more calibrated responses to outbreaks, and prevent unwanted consequences.
https://dgit.in/mathdis

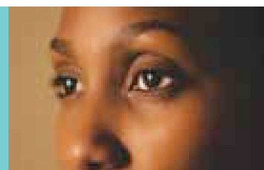


Are animals ageing gracefully?

Age related deterioration is known as "senescence". A team of researchers at ANU are attempting to figure out how the process affects various species of animals and birds. The research shows that there is significant difference between individuals in every species.
https://dgit.in/animage

Microbreweries to monitor radiation damage

Researchers have developed wearable microbreweries in disposable badges made of paper. The yeast within reacts with water to show the amount of radiation exposure. The badges can monitor radiation damage for workers in nuclear plants.
https://dgit.in/mbb



Men and women see motion differently

A study of adult men and women showed that women took between 25-75% longer to report the direction in which black and white bars were moving. There is very little evidence for difference between the sexes over low level visual processing so far.
https://dgit.in/wuvis